

Unit 2 Notes

Logic

Propositions, predicates,
reasoning, and proof

These notes develop the ideas from their basic meanings. Each section connects ordinary language to precise notation, explains a method, and includes a question with a worked solution. Diagrams use black outlines on white backgrounds so that the document can be printed without colour.

The coverage follows the supplied syllabus: prerequisite propositional logic, satisfiability, tautology, contradiction, validity, normal forms, predicates and quantifiers, nested quantifiers, rules of inference, proof techniques, mathematical induction, and answer extraction.

There are 52 numbered teaching sections, with extra topic-wise practice banks and explained answers. Read the method, follow the worked example, then attempt the questions before checking the solutions. Use the exam method sheet for revision.

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1 How to read a logical statement

Logic studies what follows from what. Its central task is not to decide whether a sentence sounds plausible. It is to identify exactly what the sentence asserts and whether a proposed conclusion is forced by the assumptions.

Suppose a rule says, “Every registered student may enter the library.” If Mira is registered, the rule supports the conclusion that Mira may enter. If Mira may enter, however, we cannot conclude that she is registered: staff members might also have permission. This difference between a rule and its reversal appears throughout the unit.

Main idea. Separate three things: the *statement*, the *assumptions*, and the *reasoning*. A statement can be true even when someone gives a poor argument for it. An argument can have a valid structure even when one of its assumptions is false.

We use letters such as p and q for whole statements. Later, expressions such as $P(x)$ describe properties of an object x . These are different levels of description: p has a truth value once its meaning is fixed; $P(x)$ may still require a value for x .

Worked example. Let p mean “the file is saved” and q mean “the backup exists.” The sentence “If the file is saved, then the backup exists” is written $p \rightarrow q$. The arrow records a conditional relationship. It does not, by itself, assert p or assert q .

Practice. A rule says, “Every square is a rectangle.” A shape is a rectangle. Does the rule establish that it is a square?

Solution. No. A rectangle with side lengths 2 and 5 satisfies the description “rectangle” but is not a square. The rule moves from being a square to being a rectangle. It does not move in the reverse direction. The example also shows how a single well-chosen counterexample can expose an invalid inference.

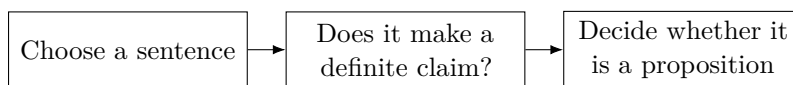
2 Propositions and truth values

A *proposition* is a declarative statement that has a definite truth value: true or false. We work in classical two-valued logic, so a proposition has exactly one of these values in a fixed interpretation.

“Seven is prime” is a proposition. “Close the door” is an instruction and has no truth value. “Is the door closed?” is a question. “ $x > 3$ ” is not yet a proposition if x has not been assigned a value or bound by a quantifier.

Context matters. “It is raining” becomes definite only after the relevant place and time are understood. Unknown truth is different from absent truth: a historical statement may be a proposition even if we do not know whether it is true.

Main idea. An *atomic proposition* is treated as a single unit. A *compound proposition* combines statements with connectives such as “and,” “or,” and “not.” Atomic does not mean short; it means that we are not analysing its internal logical structure at the chosen level.



A question or command does not have a truth value.

Worked example. Let p be “8 is even” and q be “8 is prime.” Then p is true and q is false. The compound statement “8 is even and prime” is false because its second requirement fails.

Practice. Classify: (a) 12 is divisible by 3; (b) find $12/3$; (c) n is divisible by 3; (d) every integer divisible by 6 is divisible by 3.

Solution. (a) is a true proposition. (b) is a request, not a proposition. (c) is an open statement until the meaning of n is supplied. (d) is a true proposition: the word “every” binds the variable implicitly. If an integer equals $6k$, it also equals $3(2k)$, establishing the claimed divisibility.

Common error. Do not classify a sentence as “not a proposition” merely because proving it would be difficult. Logical status and difficulty are separate issues.

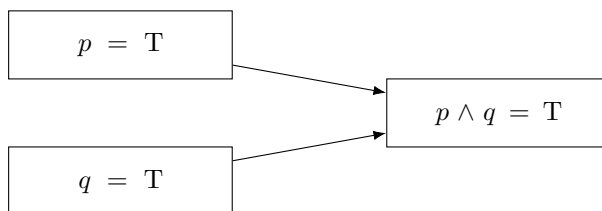
3 Negation and conjunction

The negation of p is written $\neg p$ and read “not p .” It reverses the truth value. The conjunction $p \wedge q$ means “ p and q ” and is true only when both statements are true.

p	q	$\neg p$	$p \wedge q$
T	T	F	T
T	F	F	F
F	T	T	F
F	F	T	F

Conjunction collects requirements. If access requires a valid card *and* a correct PIN, satisfying only one requirement is insufficient. In mathematics, “ x is positive and even” similarly imposes two independent conditions.

Main idea. Negation must deny the entire assertion, not replace it with a stronger assertion. The negation of “the score is greater than 50” is “the score is at most 50.” Saying “the score is less than 50” incorrectly excludes the boundary case 50.



AND needs both inputs to be true.

Worked example. If p is true and q is false, then $p \wedge \neg q$ is true: the first part is true and the negation of the false second statement is also true. But $\neg(p \wedge q)$ is also true for a different structural reason: the conjunction inside the parentheses is false.

Practice. Let p mean $x \geq 2$ and q mean $x < 7$. Translate $p \wedge q$ and $\neg p$, assuming x is real.

Solution. The conjunction means $2 \leq x < 7$: both inequalities must hold at once. The negation of p is $x < 2$. Equality belongs to the original statement and therefore does not belong to its negation. At $x = 2$, the conjunction is true; at $x = 7$, it is false.

4 Inclusive and exclusive disjunction

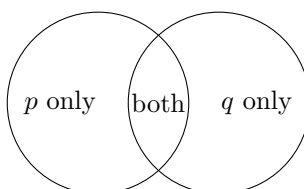
The disjunction $p \vee q$ means “ p or q , or both.” It is false only when both alternatives are false. This inclusive meaning is the default mathematical interpretation of “or.”

Exclusive or, written here as $p \oplus q$, means that exactly one alternative is true. It can be expressed using the basic connectives:

$$p \oplus q \equiv (p \vee q) \wedge \neg(p \wedge q).$$

The symbol $\equiv$ means that the expressions have the same truth value in every assignment. The first part requires at least one true alternative; the second excludes the case in which both are true.

p	q	$p \vee q$	$p \oplus q$
T	T	T	F
T	F	T	T
F	T	T	T
F	F	F	F



Inclusive OR accepts the left, middle, or right region.

Worked example. “An integer is divisible by 2 or by 3” includes 6, because 6 is divisible by both. If exactly one divisibility condition is intended, that qualification must be stated.

Practice. A form accepts a passport or a driving licence. A person supplies both. Does $p \vee q$ describe a rule that accepts this submission? Does $p \oplus q$?

Solution. The inclusive rule accepts it because at least one accepted document has been supplied. The exclusive rule rejects it because both conditions are true. The two models differ in exactly one truth-table row, but that row changes the practical meaning of the policy.

Common error. Everyday language sometimes uses “or” exclusively. Translate the intended condition rather than assuming every English “or” has the same interpretation.

5 Implication and the meaning of a conditional

The implication $p \rightarrow q$ says, “If p , then q .” Here p is the *antecedent* and q is the *consequent*. The conditional forbids one situation: p being true while q is false.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

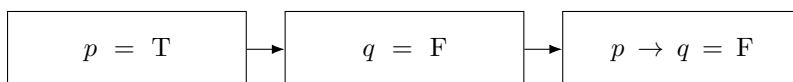
Why are the last two rows true? Consider the rule “If an application is approved, a confirmation is issued.” A rejected application does not violate this particular rule, regardless of whether some other message is issued. The rule makes a demand only in the approved case.

This is *material implication*. It does not automatically express causation, timing, or relevance. Additional assumptions are needed if those ideas matter.

Main idea. The equivalent form

$$p \rightarrow q \equiv \neg p \vee q$$

is useful for calculations. Either the triggering condition does not occur, or the required consequence holds.



This is the only false case of an implication.

Practice. Negate “If the battery is charged, the device starts.” Use p for “charged” and q for “starts.”

Solution. A failed conditional requires the trigger and a failed consequence, so the negation is $p \wedge \neg q$. In words, “The battery is charged and the device does not start.” Merely saying “the device does not start” is insufficient: perhaps the battery is not charged, in which case the original conditional has not been violated.

6 Converse, contrapositive, and biconditional

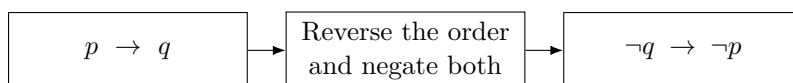
For $p \rightarrow q$, the *converse* is $q \rightarrow p$, the *inverse* is $\neg p \rightarrow \neg q$, and the *contrapositive* is $\neg q \rightarrow \neg p$. The original statement is equivalent to its contrapositive. The converse and inverse are equivalent to each other, but usually not to the original.

The biconditional $p \leftrightarrow q$ means “ p if and only if q .” It requires both directions:

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p).$$

It is true when p and q have the same truth value, including when both are false.

Main idea. In $p \rightarrow q$, p is *sufficient* for q , and q is *necessary* for p . “ p only if q ” therefore means $p \rightarrow q$. “ p if q ” means $q \rightarrow p$.



The contrapositive keeps the meaning of the original implication.

Worked example. “An integer is divisible by 4 only if it is even” means $D_4 \rightarrow E$. Divisibility by 4 is enough to ensure evenness; evenness is required for divisibility by 4. The converse fails at 6.

Practice. Translate “Access is granted if and only if the account is active.” Then state its two separate obligations.

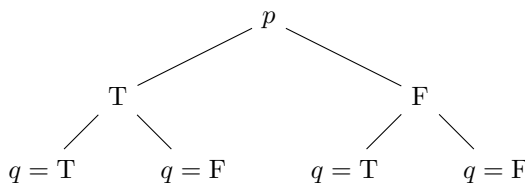
Solution. With a for access and b for an active account, the translation is $a \leftrightarrow b$. The obligations are $a \rightarrow b$ and $b \rightarrow a$. The first prohibits access for an inactive account. The second prohibits denying access to an active account. A rule that enforces only the first condition is weaker than the biconditional.

Common error. When proving an “if and only if” theorem, label and prove both directions. A correct proof of one implication completes only half the task.

7 Constructing a truth table systematically

A truth table lists every possible assignment of truth values to the atomic propositions. With n distinct atoms there are 2^n rows, because each atom independently has two choices. Repeated occurrences of the same atom do not create new choices.

Use a fixed row order. For three atoms, let p stay true for four rows, let q change every two rows, and let r change every row. This prevents missing or duplicating an assignment. Evaluate inside parentheses first and create intermediate columns whenever the expression becomes hard to read.



Two choices for each of two variables give four rows.

Worked example. Examine $(p \rightarrow q) \wedge p$:

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge p$
T	T	T	T
T	F	F	F
F	T	T	F
F	F	T	F

The final column is true only in the first row. In that row q is also true. This observation is the truth-table basis of modus ponens, a rule introduced later.

Practice. How many rows are needed for $(p \wedge q) \vee (\neg p \wedge r)$? Evaluate it when $p = F$, $q = F$, and $r = T$.

Solution. There are three distinct atoms, so eight rows are needed. At the specified assignment, $p \wedge q$ is false, while $\neg p \wedge r$ is true. Their disjunction is true. Notice that q is irrelevant on this particular branch because p is false; nevertheless, all three atoms still belong in the full truth table.

Common error. A partially completed table can prove satisfiability if it finds a true row. It cannot prove tautology unless every possible row has been checked or another complete argument is supplied.

8 Equivalence and the basic algebra of logic

Two formulas are *logically equivalent* if they agree under every truth assignment. We write $A \equiv B$. This is a relationship between formulas. The formula $A \leftrightarrow B$ is a tautology exactly when that relationship holds.

The following laws work for arbitrary formulas, not just single letters. Here T and F denote constant truth and falsehood.

Law	Equivalent forms	
Identity	$p \wedge T \equiv p$	$p \vee F \equiv p$
Domination	$p \vee T \equiv T$	$p \wedge F \equiv F$
Idempotence	$p \wedge p \equiv p$	$p \vee p \equiv p$
Double negation	$\neg\neg p \equiv p$	
Complement	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$

Conjunction and disjunction are commutative, so the order of their inputs can be exchanged. They are also associative, so a chain of the same connective can be regrouped without changing its value.

Worked example. Simplify $(p \wedge T) \vee (p \wedge F)$. The left term becomes p by identity. The right term becomes F by domination. The resulting $p \vee F$ is p .

Practice. Simplify $\neg\neg q \wedge (q \vee \neg q)$, naming the main steps.

Solution. Double negation changes the first factor to q . The complement law changes the second factor to T. Identity then gives $q \wedge T \equiv q$. Each replacement is valid inside the larger expression because equivalent subformulas are interchangeable.

Common error. Logical algebra resembles numerical algebra but has different laws. For example, $p \vee p \equiv p$ has no ordinary numerical analogue $x + x = x$. Use logical identities explicitly rather than transferring arithmetic habits without checking them.

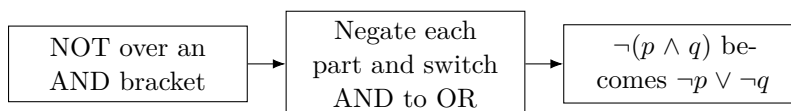
9 De Morgan's laws and negating compound claims

De Morgan's laws explain how negation passes through “and” and “or”:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q,$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q.$$

The connective changes when the negation moves inward. To deny that both requirements hold, at least one must fail. To deny that either alternative holds, both must fail.



Apply the change to the whole bracket.

Worked example. Suppose successful login requires a valid password and an active account. The denial of successful satisfaction of these requirements is “the password is invalid or the account is inactive.” It need not be the case that both fail.

For three or more inputs, the same reasoning applies. The negation of $p \wedge q \wedge r$ is $\neg p \vee \neg q \vee \neg r$. Parentheses make the scope clear when different connectives occur together.

Practice. Simplify $\neg((p \vee q) \wedge \neg r)$ until negations occur only directly before atoms.

Solution. First apply De Morgan's law to the outer conjunction:

$$\neg(p \vee q) \vee \neg\neg r.$$

Next negate the inner disjunction and remove the double negation:

$$(\neg p \wedge \neg q) \vee r.$$

In words, the original requirement fails if both p and q fail, or if r holds. This explanation is useful for checking that the symbols retain the intended meaning.

Common error. Writing $\neg(p \vee q)$ as $\neg p \vee \neg q$ is wrong. If p is true and q is false, the first expression is false and the proposed replacement is true. A single disagreeing row disproves equivalence.

10 Distributive and absorption laws

Both connectives distribute over the other:

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r),$$

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r).$$

The first law separates two ways of satisfying a shared requirement. The second law is just as valid, even though it feels less familiar from ordinary algebra.

Absorption removes a condition already covered by another:

$$p \vee (p \wedge q) \equiv p,$$

$$p \wedge (p \vee q) \equiv p.$$

For the first law, if p is true the whole expression is true; if p is false both alternatives are false. The value of q never changes the outcome.

Worked example. Simplify $(p \wedge q) \vee (p \wedge \neg q)$. Factor the common requirement p using distributivity:

$$p \wedge (q \vee \neg q) \equiv p \wedge \mathbf{T} \equiv p.$$

The expression describes two alternatives that together cover every possibility for q , while both require p .

Practice. Simplify $(p \vee q) \wedge (p \vee \neg q)$.

Solution. Use the second distributive law in reverse:

$$(p \vee q) \wedge (p \vee \neg q) \equiv p \vee (q \wedge \neg q)$$

$$\equiv p \vee \mathbf{F}$$

$$\equiv p.$$

To check the meaning directly, suppose p is false. Then the two parentheses demand both q and $\neg q$, which is impossible. If p is true, both parentheses hold regardless of q .

Common error. Distributing in the wrong direction can make a problem much longer. Decide whether your goal is simplification, a particular normal form, or comparison with another formula before expanding everything.

Practice bank: Connectives and logical laws

Attempt these questions before reading the answers on the following page. For a calculation or proof, write the method and the essential working as well as the final result.

1. If $p = \text{T}$, $q = \text{F}$, find $\neg p$, $p \wedge q$, and $p \vee q$.
2. With $p = \text{F}$, $q = \text{T}$, find $p \rightarrow q$ and $p \leftrightarrow q$.
3. Negate $x > 7$, with real numbers as the domain.
4. Simplify $p \wedge (p \vee q)$.
5. Write $p \rightarrow (q \vee r)$ without an implication sign.
6. Write the converse and contrapositive of $p \rightarrow q$.
7. Simplify $\neg(p \wedge \neg q)$.
8. Translate: access is permitted only if the card is valid.

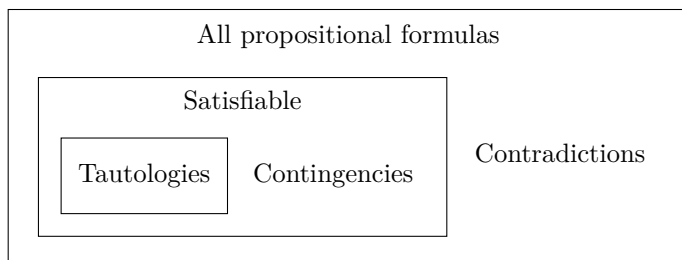
Answers: Connectives and logical laws

Compare the reasoning as well as the final answer. If your answer differs, return to the first step where the methods separate.

1. The answers are F, F, T. NOT reverses p ; AND requires both statements; OR requires at least one.
2. The conditional is true because its antecedent is false. The biconditional is false because the values differ.
3. The negation is $x \leq 7$. Include equality because $x = 7$ does not satisfy the original strict inequality.
4. The answer is p , by absorption. If p is true both requirements hold; if p is false the conjunction fails.
5. It becomes $\neg p \vee q \vee r$. Replace the antecedent by its negation and join it to the consequent with OR.
6. The converse is $q \rightarrow p$. The contrapositive is $\neg q \rightarrow \neg p$. Only the contrapositive is always equivalent to the original.
7. De Morgan's law gives $\neg p \vee \neg \neg q$. Remove double negation to obtain $\neg p \vee q$.
8. Let a mean access is permitted and c mean the card is valid. The answer is $a \rightarrow c$. The card condition is necessary for permission.

11 Satisfiability, tautology, and contradiction

A formula is *satisfiable* when at least one truth assignment makes it true. A satisfying assignment is also called a *model*. A *tautology* is true in every assignment. A *contradiction* is false in every assignment. A *contingent* formula is true in some assignments and false in others.



Every tautology is satisfiable, but many satisfiable formulas are not tautologies. A contradiction is also called *unsatisfiable*. These definitions concern all possible assignments, not just the situation currently under discussion.

Worked example. $p \vee \neg p$ is a tautology. $p \wedge \neg p$ is a contradiction. $p \rightarrow q$ is contingent: it is false at $p = \text{T}, q = \text{F}$ and true in the other rows.

Practice. Classify $(p \vee q) \wedge \neg p$. Give evidence for your classification.

Solution. It is contingent. At $p = \text{F}, q = \text{T}$, both the disjunction and $\neg p$ are true. At $p = \text{T}, q = \text{T}$, the second factor is false. One true row establishes satisfiability; one false row rules out tautology. Together they establish contingency.

Common error. “Not a tautology” does not mean “a contradiction.” There is a large middle category: formulas whose values depend on the assignment.

12 Checking whether an argument is valid

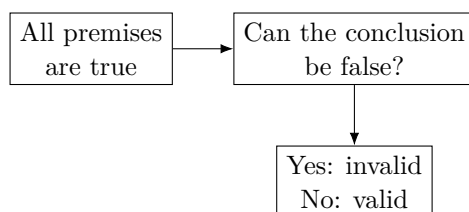
An argument gives some statements, called premises, and asks you to accept a conclusion. A valid argument is one in which the conclusion must be true whenever all the premises are true.

Truth-table method.

1. Make columns for the premises and the conclusion.
2. Find the rows where every premise is true.
3. In those rows, check the conclusion. If it is always true, the argument is valid.
4. If even one of those rows has a false conclusion, the argument is invalid.

Rows containing a false premise do not disprove validity. You are checking what follows when the premises all hold.

Worked example. Test: $p \rightarrow q$, p ; therefore q . When p is true, the first premise can remain true only if q is true. Thus there is no row with true premises and a false conclusion. The argument is valid.



Practice. Test: $p \rightarrow q$, q ; therefore p .

Solution. Set $p = F$, $q = T$. Both premises are true, but the conclusion is false. Hence the argument is invalid. This row is called a counterexample or countermodel.

Notation you may see. $P_1, P_2 \models C$ means that the premises P_1, P_2 imply the conclusion C . The symbol $\models$ is simply a shorthand for the validity relationship.

Common error. A valid argument can start from an untrue premise. If the argument is valid and its premises are actually true, it is called sound.

13 Finding one truth assignment that works

A truth assignment means choosing true or false for each letter. To show that a formula is satisfiable, you need only one assignment that makes the whole formula true.

Practical method. Start with a part that gives a definite instruction. For example, $\neg q$ requires q to be false. Then use that choice to simplify the other parts.

Worked example. Find values satisfying

$$(p \vee q) \wedge (\neg p \vee r) \wedge \neg q.$$

1. The last part, $\neg q$, gives $q = \text{F}$.
2. The first part is now $p \vee \text{F}$, so $p = \text{T}$.
3. The middle part is now $\text{F} \vee r$, so $r = \text{T}$.

Thus one answer is $p = \text{T}, q = \text{F}, r = \text{T}$. Put these values back into every original bracket to check the answer.

For an invalid argument, find values that make all its premises true and its conclusion false. Start by making the conclusion false, then see whether the premises can still work.

Practice. Do $p \vee q$ and $p \rightarrow r$ force r to be true?

Solution. Try $r = \text{F}$. For $p \rightarrow r$ to hold, use $p = \text{F}$. Then $p \vee q$ requires $q = \text{T}$. The values $p = \text{F}, q = \text{T}, r = \text{F}$ make both premises true and the conclusion false. The argument is invalid.

Practice. Can $p \wedge \neg p$ ever be true?

Solution. No. If p is true, $\neg p$ is false. If p is false, the first part is false. In both cases the conjunction fails.

Common error. If one guessed assignment fails, try another. One failed attempt does not show that every possible assignment fails.

14 Literals, clauses, and normal forms

A *literal* is an atom or its negation: p and $\neg p$ are literals, but $\neg(p \vee q)$ is not. A *clause* is a disjunction of literals, such as $p \vee \neg q \vee r$. A *term*, in the propositional normal-form discussion, is a conjunction of literals.

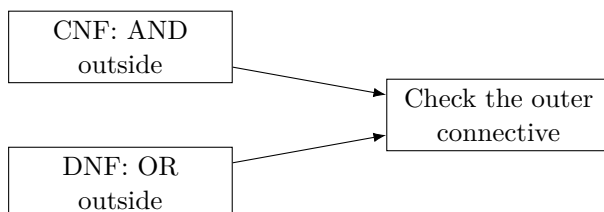
A formula is in *conjunctive normal form* (CNF) if it is a conjunction of clauses. It is in *disjunctive normal form* (DNF) if it is a disjunction of terms. The outside connective supplies the name.

$$(p \vee \neg q) \wedge (q \vee r) \quad \text{is CNF,}$$

$$(p \wedge \neg q) \vee (q \wedge r) \quad \text{is DNF.}$$

A single literal can count as a one-literal clause or term. Thus $p \wedge q$ is both CNF, as a conjunction of singleton clauses, and DNF, as one conjunctive term.

Main idea. CNF describes simultaneous constraints: every clause must be satisfied. DNF describes alternative successful cases: at least one term must be satisfied. This difference explains why each form is useful for different tasks.



The outer connective identifies the normal form.

Worked example. $(p \rightarrow q) \wedge r$ is not yet in CNF because an implication remains. Replacing the conditional with $\neg p \vee q$ gives $(\neg p \vee q) \wedge r$, which is CNF.

Practice. Classify $(p \wedge q) \vee \neg r$ and $\neg(p \wedge q) \vee r$ as written.

Solution. The first is DNF: it is a disjunction of the term $p \wedge q$ and the singleton term $\neg r$. The second is neither of the stated normal forms as written, because a negation covers a compound expression. After De Morgan's law it becomes $\neg p \vee \neg q \vee r$, a single clause and hence CNF. It is also DNF if each literal is viewed as a singleton term.

15 CNF: the steps to follow

CNF means *conjunctive normal form*. Think of it as several OR brackets joined by AND:

$$(\text{literal OR literal}) \text{ AND } (\text{literal OR literal}).$$

A literal is a letter or its negation, such as p or $\neg p$. A bracket may contain just one literal.

Conversion recipe.

1. Remove arrows: replace $p \rightarrow q$ with $\neg p \vee q$.
2. Push NOT signs inward using De Morgan's laws.
3. For CNF, expand OR over AND:

$$A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C).$$

4. Check that the outside connective is AND and that each bracket contains only ORs of literals.

Worked example. Convert $p \rightarrow (q \wedge r)$ into CNF. Removing the arrow gives $\neg p \vee (q \wedge r)$. Expand:

$$(\neg p \vee q) \wedge (\neg p \vee r).$$

This is already CNF. The two brackets are joined by AND.

Practice. Convert $(p \wedge q) \rightarrow r$ into CNF.

Solution. Remove the arrow to get $\neg(p \wedge q) \vee r$. De Morgan's law gives $\neg p \vee \neg q \vee r$. A single OR bracket is allowed, so this is CNF.

Practice. Convert $p \vee (q \wedge r)$ into CNF.

Solution. Use the expansion rule directly: $(p \vee q) \wedge (p \vee r)$.

Common error. Do not expand in the opposite direction by habit. CNF wants AND on the outside. DNF, on the next page, wants OR on the outside.

16 DNF: the steps to follow

DNF means *disjunctive normal form*. It is a collection of AND groups joined by OR:

$$(\text{literal AND literal}) \text{ OR } (\text{literal AND literal}).$$

The letters in a group may be positive or negated. Each group describes one way the formula can be true.

Conversion recipe. Remove arrows, push NOT signs inward, then expand AND over OR:

$$A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C).$$

After expansion, remove an impossible group such as $p \wedge \neg p$ if one appears.

Worked example. Convert $(p \vee q) \wedge \neg r$ into DNF. Attach the common $\neg r$ to each alternative:

$$(p \wedge \neg r) \vee (q \wedge \neg r).$$

The outside connective is OR, so the required shape is obtained.

Practice. Find a DNF for $(p \vee q) \wedge r$.

Solution. Distribute r over the two alternatives:

$$(p \wedge r) \vee (q \wedge r).$$

Practice. Find a DNF for $(p \vee q) \wedge (\neg p \vee r)$.

Solution. Multiply out the brackets using logical AND:

$$\begin{aligned} & (p \wedge \neg p) \vee (p \wedge r) \\ & \vee (q \wedge \neg p) \vee (q \wedge r). \end{aligned}$$

Remove the impossible first group. A correct DNF is

$$(p \wedge r) \vee (q \wedge \neg p) \vee (q \wedge r).$$

This answer is sufficient if the question asks for a DNF; further minimisation is a separate task.

Common error. An OR group is not the same as an AND group. Read your final answer's outer connective to check that you have produced the requested form.

17 Principal CNF and DNF from a truth table

Principal forms use every variable in every group. They are also called canonical forms. The truth table gives a direct recipe.

Principal DNF: use the true rows. For each true row, join the variables with AND. A variable that is true stays positive; a variable that is false gets a NOT sign. Join the resulting groups with OR.

Principal CNF: use the false rows. For each false row, join the variables with OR. Reverse the signs: a true variable gets NOT, while a false variable stays positive. Join these brackets with AND.

Row	DNF group matching it	CNF bracket excluding it
$p = \text{T}, q = \text{F}$	$p \wedge \neg q$	$\neg p \vee q$
$p = \text{F}, q = \text{T}$	$\neg p \wedge q$	$p \vee \neg q$

Worked example. For exclusive OR, the true rows are (T, F) and (F, T). Hence its principal DNF is

$$(p \wedge \neg q) \vee (\neg p \wedge q).$$

The false rows are (T, T) and (F, F). Hence its principal CNF is $(\neg p \vee \neg q) \wedge (p \vee q)$.

Practice. Find the principal CNF of $p \rightarrow q$.

Solution. The only false row is $p = \text{T}, q = \text{F}$. Reverse the signs and join with OR: $\neg p \vee q$.

Practice. Find its principal DNF.

Solution. The other three rows are true. Write one matching AND group for each:

$$(p \wedge q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q).$$

Common error. A DNF group must be true in its selected row. A CNF bracket must be false in its selected row. Testing this once is an easy way to check your signs.

18 Review: propositional reasoning in one problem

Consider a system with three statements: p means “permission is granted,” q means “the account is verified,” and r means “a supervisor approves.” The policy is

$$F = (p \rightarrow q) \wedge (p \rightarrow r) \wedge ((q \wedge r) \rightarrow p).$$

Read it before manipulating it. Permission requires verification and approval. Conversely, having both verification and approval is enough for permission. Together, the three requirements say that permission is granted exactly when both conditions hold.

Worked example. In compact form, $F \equiv p \leftrightarrow (q \wedge r)$. Eliminating the arrows gives the CNF

$$(\neg p \vee q) \wedge (\neg p \vee r) \wedge (\neg q \vee \neg r \vee p).$$

This form makes each prohibited combination explicit. The first clause rules out permission without verification, the second permission without approval, and the third denial of permission when both are present.

Practice. (a) Is F satisfiable? (b) Is F a tautology? (c) Do F and q entail p ? (d) Do F, q, r entail p ?

Solution. (a) Yes. The assignment $p = q = r = \text{T}$ satisfies every clause. (b) No. The assignment $p = \text{T}, q = \text{F}, r = \text{T}$ violates the first clause. Hence F is contingent.

(c) No. Set $q = \text{T}, r = \text{F}, p = \text{F}$. The policy holds, and q holds, but permission is false because approval is missing. (d) Yes. The third implication now has a true antecedent, so p must be true.

This example combines translation, equivalence, normal forms, models, and validity. In a longer question, handle these as separate steps. A correct translation gives the later algebra a meaningful target.

Practice bank: Truth tables, normal forms, and validity

Attempt these questions before reading the answers on the following page. For a calculation or proof, write the method and the essential working as well as the final result.

1. How many rows are needed for a truth table containing four distinct variables?
2. Classify $p \vee \neg p$.
3. Classify $p \wedge \neg p$.
4. Give one satisfying assignment for $(p \vee q) \wedge \neg p$.
5. Find a CNF for $p \vee (q \wedge r)$.
6. Find a DNF for $p \wedge (q \vee r)$.
7. Do $p \rightarrow q$ and $\neg q$ imply $\neg p$?
8. Do $p \rightarrow q$ and $\neg p$ imply $\neg q$? Give a test assignment.

Answers: Truth tables, normal forms, and validity

Compare the reasoning as well as the final answer. If your answer differs, return to the first step where the methods separate.

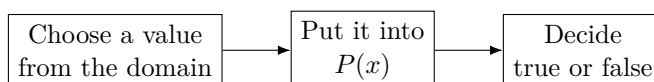
1. $2^4 = 16$. Each variable has two possible truth values.
2. It is a tautology because it is true for both possible values of p .
3. It is a contradiction because its two parts can never be true together.
4. Use $p = F, q = T$. The OR bracket and the NOT condition are both true.
5. Distribute OR over AND to get $(p \vee q) \wedge (p \vee r)$.
6. Distribute AND over OR to get $(p \wedge q) \vee (p \wedge r)$.
7. Yes, by modus tollens. A true p would require a true q , so the known false q rules out p .
8. No. Set $p = F, q = T$. Both premises are true and the proposed conclusion is false.

19 Predicates: a statement with a blank

A predicate is a statement containing a variable. Think of the variable as a blank waiting for a value. For example, let $E(x)$ mean that x is even. Then $E(4)$ is true and $E(5)$ is false.

The domain tells you which values may fill the blank. It could be integers, real numbers, students, or something else. Write down the domain when the question does not make it obvious.

Worked example. Let $P(x)$ mean $x^2 = 4$, with integers as the domain. Then $P(2)$ and $P(-2)$ are true. The statement $P(3)$ is false because $3^2 = 9$.



A predicate can contain two variables. For example, $L(x, y)$ may mean $x < y$. Then $L(2, 5)$ is true and $L(5, 2)$ is false. The order of the inputs matters.

Practice. Let $D(x, y)$ mean that x divides y . Decide $D(3, 12)$ and $D(12, 3)$.

Solution. The first is true because $12 = 3 \cdot 4$. The second is false because no integer multiple of 12 equals 3.

Practice. Why must the domain be stated for $x^2 = 2$?

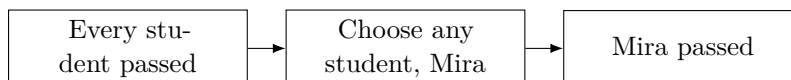
Solution. It has a solution among real numbers, such as $\sqrt{2}$, but no solution among integers. Changing the allowed values changes the answer.

Common error. $P(x)$ by itself may still contain an unspecified value. A quantifier, such as every or some, turns it into a statement about the domain.

20 Universal quantification

The statement $\forall x P(x)$ means that $P(x)$ holds for every object in the declared domain. The symbol $\forall$ is the universal quantifier. The variable x is bound by it, and the formula following the quantifier is its scope.

To establish a universal claim, reason about an arbitrary domain element without relying on special properties not shared by all elements. To refute it, provide one domain element for which the predicate fails. Such an element is a *counterexample*.



A universal claim applies to each permitted example.

Worked example. Over integers, $\forall x (x^2 \geq 0)$ is true. Over integers, $\forall x (x^2 > x)$ is false: choose $x = 0$ or $x = 1$. Testing $x = 2, 3, 4$ would not have justified the universal statement.

On a finite domain $\{a, b, c\}$, universal quantification is equivalent to a conjunction:

$$\forall x P(x) \equiv P(a) \wedge P(b) \wedge P(c).$$

On an infinite domain, a finite sample cannot replace the quantifier.

Practice. Translate “Every student in the class submitted the assignment.” First use a domain consisting only of class members, then use a domain of all people.

Solution. On the restricted domain, write $\forall x S(x)$, where $S(x)$ means that x submitted. On the larger domain, let $C(x)$ mean membership in the class and write

$$\forall x (C(x) \rightarrow S(x)).$$

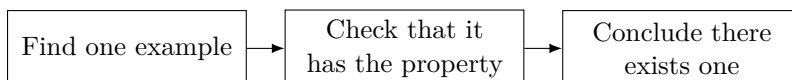
The implication limits the requirement to class members. Replacing it with $C(x) \wedge S(x)$ would claim that every person belongs to the class as well as having submitted, which is much stronger.

Common error. A universal claim over a category with no members can be true vacuously. “Every absent student submitted” imposes no violated condition if there are no absent students. It does not establish that an absent student exists.

21 Existential quantification and witnesses

The statement $\exists x P(x)$ means that at least one domain element satisfies P . The symbol $\exists$ is the existential quantifier. An element making the predicate true is called a *witness*.

An existential statement does not normally claim uniqueness. “There exists a solution” allows two, ten, or infinitely many solutions. If exactly one is intended, existence and uniqueness must both be stated.



One suitable example establishes an existential claim.

Worked example. Over integers, $\exists x (x^2 = 9)$ is true because $x = 3$ is a witness. The value -3 is another witness. Over integers, $\exists x (x^2 = 2)$ is false; over reals, it is true. The quantifier cannot be interpreted independently of the domain.

On the finite domain $\{a, b, c\}$,

$$\exists x P(x) \equiv P(a) \vee P(b) \vee P(c).$$

This is the existential counterpart of the universal conjunction.

Practice. Translate “Some student passed every required examination,” using people as the domain for x and examinations as the domain for y . Let $S(x)$ mean student, $R(y)$ mean required, and $P(x, y)$ mean passed.

Solution. Write

$$\exists x (S(x) \wedge \forall y (R(y) \rightarrow P(x, y))).$$

The conjunction ensures that the witness really is a student. The inner implication makes a passing requirement only for required examinations. Using $S(x) \rightarrow \dots$ at the outer level would allow a nonstudent to satisfy the existential statement merely by making its antecedent false.

Common error. When proving existence, show both that the proposed witness belongs to the domain and that it satisfies the whole required property. Naming an object without checking its conditions is an incomplete proof.

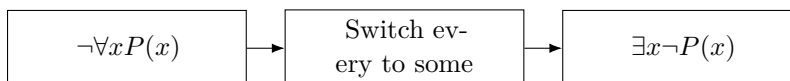
22 Negating quantified statements

To deny “every object has the property,” it is enough to find one object without it. To deny “some object has the property,” every object must lack it. Therefore

$$\neg \forall x P(x) \equiv \exists x \neg P(x),$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x).$$

These are the quantifier versions of De Morgan’s laws. Each quantifier changes when a negation passes through it.



Denying every requires one exception.

Worked example. Negate “Every student passed at least one test.” A formal version is $\forall x \exists y P(x, y)$ when the domains already consist of students and tests. Moving the negation inward gives

$$\exists x \forall y \neg P(x, y).$$

In words, “There is a student who passed no test.” This is stronger than saying that some student failed some test; the original statement permits failed tests as long as each student passed at least one.

Practice. Over real numbers, negate $\forall x \exists y (y > x \wedge y < x + 1)$.

Solution. First switch the universal quantifier, then the existential one, then negate the conjunction:

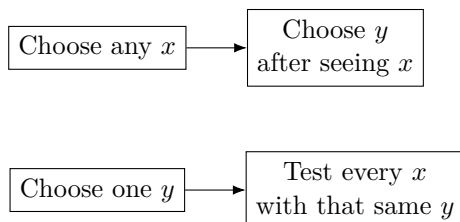
$$\exists x \forall y (y \leq x \vee y \geq x + 1).$$

The result says that some open interval $(x, x + 1)$ contains no real number. It is false, since $y = x + \frac{1}{2}$ lies in every such interval. The negation’s falsity is consistent with the original statement’s truth.

Common error. After switching quantifiers, remember to negate the predicate itself. Also preserve boundary cases: the negation of $y > x$ is $y \leq x$, not $y < x$.

23 Nested quantifiers and order of choice

Mixed quantifiers have an order that controls dependence. In $\forall x \exists y R(x, y)$, the witness y may depend on x . In $\exists y \forall x R(x, y)$, one fixed y must work for every x .



Worked example. Over integers, $\forall x \exists y (y > x)$ is true: after receiving x , choose $y = x + 1$. But $\exists y \forall x (y > x)$ is false. For any fixed candidate y , the choice $x = y$ defeats the strict inequality.

Quantifiers of the same kind commute: $\forall x \forall y$ can be swapped, and $\exists x \exists y$ can be swapped. Mixed quantifiers cannot generally be swapped. The statement with a single uniform witness, $\exists y \forall x R(x, y)$, does imply $\forall x \exists y R(x, y)$, but the reverse implication need not hold.

Practice. Compare “Every book has an author” with “There is a person who authored every book.”

Solution. The first is $\forall b \exists a A(a, b)$; different books may have different authors. The second is $\exists a \forall b A(a, b)$; one person must qualify for every book. Two books with different sole authors provide a situation in which the first statement is true and the second false.

24 Relations as tables: seeing quantifier order

A finite relation can be displayed as a table. Rows represent objects chosen for the first variable and columns represent objects chosen for the second. A 1 means that the relation holds; a 0 means that it does not.

$R(x, y)$	$y = a$	$y = b$	$y = c$
$x = 1$	1	0	0
$x = 2$	0	1	0
$x = 3$	0	0	1

Here $\forall x \exists y R(x, y)$ is true because each row contains a 1. But $\exists y \forall x R(x, y)$ is false because no column consists entirely of 1s. The table gives a concrete counterexample to swapping mixed quantifiers.

The statement $\exists x \forall y R(x, y)$ asks for a row entirely of 1s. The statement $\forall y \exists x R(x, y)$ asks for at least one 1 in every column. Read the outer quantifier first, then translate the inner requirement into a row or column condition.

Practice. Replace the last row by $(1, 1, 1)$. Which of the four statements in this section become true?

Solution. Every row still has a 1, so $\forall x \exists y R(x, y)$ remains true. Every column has a 1 in the last row, so $\forall y \exists x R(x, y)$ is true. There is now an all-1 row, making $\exists x \forall y R(x, y)$ true. There is still no all-1 column because the first two rows have their 1s in different columns. Thus $\exists y \forall x R(x, y)$ remains false.

This visual method is especially useful when a verbal description is confusing. Construct the smallest finite example that displays the intended dependence, then read the quantifiers directly from it.

Practice bank: Predicates and quantifiers

Attempt these questions before reading the answers on the following page. For a calculation or proof, write the method and the essential working as well as the final result.

1. Using students as the domain, write that every student passed. Let $P(x)$ mean passed.
2. Using all people as the domain, write that every student passed. Use $S(x)$ for student.
3. Using all people as the domain, write that some student passed.
4. Negate $\forall x P(x)$.
5. Negate $\exists x \forall y R(x, y)$.
6. Over integers, decide $\forall x \exists y (y = x + 2)$.
7. Over integers, decide $\exists y \forall x (y = x + 2)$.
8. Negate: every student read at least one book.

Answers: Predicates and quantifiers

Compare the reasoning as well as the final answer. If your answer differs, return to the first step where the methods separate.

1. Write $\forall xP(x)$. The domain already restricts the discussion to students.
2. Write $\forall x(S(x) \rightarrow P(x))$. The implication applies the passing requirement to students.
3. Write $\exists x(S(x) \wedge P(x))$. The chosen person must be both a student and someone who passed.
4. The negation is $\exists x\neg P(x)$: at least one object does not have the property.
5. Switch the quantifiers and negate the relation: $\forall x\exists y\neg R(x, y)$.
6. True. For each given integer x , choose the integer $y = x + 2$. The choice may change with x .
7. False. The choice $x = 0$ requires $y = 2$, while $x = 1$ requires $y = 3$. One fixed y cannot do both.
8. Some student read no books. Symbolically, the negation of $\forall x\exists yR(x, y)$ is $\exists x\forall y\neg R(x, y)$, using the stated domains.

25 Reading brackets and the scope of a quantifier

The scope of a quantifier is the part of the formula it controls. Brackets make the scope visible. Before translating a formula, identify which variables belong to each quantifier.

Worked example. Compare

$$(\forall x P(x)) \vee Q(x) \quad \text{and} \quad \forall x (P(x) \vee Q(x)).$$

In the first formula, $\forall x$ controls only $P(x)$. The x in $Q(x)$ still needs a value. In the second formula, the quantifier controls both parts of the bracket.

A variable controlled by a quantifier is called bound. A variable occurrence not controlled by one is called free. These names are useful for short exam questions, but the practical task is simply to trace each quantifier to its bracket.

Three-step method.

1. Locate a quantifier and its variable.
2. Mark the expression within its scope.
3. Identify matching variable occurrences inside that expression.

Practice. In $\forall x R(x, y)$, which variable is free?

Solution. The x is bound by $\forall x$. The y is free because no quantifier controls it.

Practice. Does changing $\forall x P(x)$ to $\forall t P(t)$ change the meaning?

Solution. No. The letter naming a bound variable can be changed consistently. Both formulas say that every domain element has property P .

Substitution caution. In $\exists y L(x, y)$, if you want to replace the free x with y , first rename the bound y to z . Then the answer is $\exists z L(y, z)$. This keeps the two roles separate.

Common error. Do not move a quantifier past brackets without checking its scope. A small change in brackets can change the whole sentence.

26 Translating exactly one and at least two

An existential quantifier means at least one. It does not mean exactly one. To write exactly one, state that a suitable object exists and that every other suitable object is the same one.

Exactly-one pattern.

$$\exists x(P(x) \wedge \forall y(P(y) \rightarrow y = x)).$$

Read this in two pieces. First, choose an x with property P . Second, any y with property P must equal that x .

Worked example. If the domain is students and $W(x)$ means that x won a prize, exactly one student won a prize is

$$\exists x(W(x) \wedge \forall y(W(y) \rightarrow y = x)).$$

It says that there is a winner, and there cannot be a different winner.

At-least-two pattern.

$$\exists x \exists y(P(x) \wedge P(y) \wedge x \neq y).$$

The condition $x \neq y$ ensures that you have two different objects.

Practice. Write that at least two students passed, using $P(x)$ for passed.

Solution. With students as the domain, write

$$\exists x \exists y(P(x) \wedge P(y) \wedge x \neq y).$$

Practice. Is $\exists x \exists y(P(x) \wedge P(y))$ enough to say two students passed?

Solution. No. Both variables could refer to the same student. Add $x \neq y$.

At most one. Use $\forall x \forall y((P(x) \wedge P(y)) \rightarrow x = y)$. This allows no suitable objects or one suitable object. Exactly one adds the existence requirement.

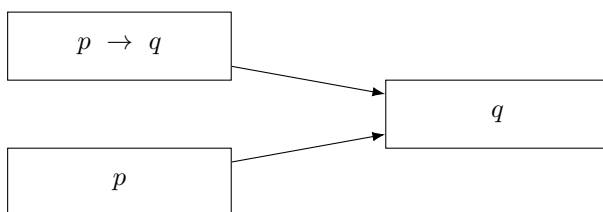
Common error. Keep the domain in mind. If it includes all people rather than just students, also include a condition saying that the chosen people are students.

27 Rules of propositional inference

A rule of inference is a small valid argument pattern that can be reused inside a larger proof. In the patterns below, p, q, r may stand for compound formulas as well as atoms.

Rule	Premises	Conclusion
Modus ponens	$p \rightarrow q, p$	q
Modus tollens	$p \rightarrow q, \neg q$	$\neg p$
Hypothetical syllogism	$p \rightarrow q, q \rightarrow r$	$p \rightarrow r$
Disjunctive syllogism	$p \vee q, \neg p$	q
Conjunction	p, q	$p \wedge q$
Simplification	$p \wedge q$	p
Addition	p	$p \vee q$

Modus ponens applies a rule when its trigger is known. Modus tollens uses a failed consequence to rule out its trigger. Addition may seem to introduce unnecessary information, but the conclusion is weaker than the premise: if p is true, at least one of p, q is certainly true.



Modus ponens applies a rule to a known true starting condition.

Worked example. From $p \rightarrow q$, $q \rightarrow r$, and $\neg r$, first obtain $\neg q$ by modus tollens on the second implication. Then obtain $\neg p$ by modus tollens on the first. Each step uses a previously available statement.

Practice. Derive s from $p \vee q$, $\neg p$, and $q \rightarrow s$.

Solution. The first two premises yield q by disjunctive syllogism. Combining q with $q \rightarrow s$ yields s by modus ponens. A clear formal proof would number the premises and cite their numbers beside each derived line. This records which information supports which conclusion.

Common error. A rule applies to a complete matching pattern. From $p \vee q$ alone, neither p nor q may be selected. An alternative is known to hold, but the premise does not say which one.

28 Invalid inference patterns and counterexamples

Two frequent errors resemble valid rules but reverse the direction of the information.

Affirming the consequent uses $p \rightarrow q$ and q to infer p . It is invalid because q may hold for another reason. *Denying the antecedent* uses $p \rightarrow q$ and $\neg p$ to infer $\neg q$. It is invalid because the conditional does not state that p is the only route to q .

Worked example. “If a number is divisible by 4, it is even. The number is even. Therefore it is divisible by 4.” The number 6 defeats the argument. For denying the antecedent, use “6 is not divisible by 4, therefore it is not even.” The same number again exposes the error.

The assignment $p = \text{F}, q = \text{T}$ is a countermodel to both patterns. In one pattern it makes the asserted p false; in the other it makes the asserted $\neg q$ false, while the premises remain true.

Practice. A student argues: “If the program is correct, it passes the test. It passes the test, so it is correct.” Identify the pattern and repair the conclusion without adding unsupported information.

Solution. This affirms the consequent. A program with an untested defect could pass the particular test. From the premises alone, correctness is not established. A justified statement is simply that passing this test is consistent with correctness. If the test had been proved sufficient for correctness, that would supply the missing converse, but it is an additional premise.

Common error. An invalid pattern can sometimes have a true conclusion. The issue is that the conclusion is not guaranteed by the given premises. Never use the accidental truth of one example to defend the general validity of an argument form.

29 Using every and some in a proof

Quantifier rules become easier when read as ordinary instructions.

From every to a named object. If every student passed and Mira is a student, you may conclude that Mira passed. This is universal instantiation.

From a named example to some. If Mira passed, you may conclude that someone passed. This is existential generalisation.

From an arbitrary object to every. To show that every integer has a property, take an arbitrary integer and establish the property without choosing a special value.

Using some. If some object has a property, give it a fresh temporary name while working. Do not assume it is a particular person already mentioned.

Worked example. Given $\forall x(S(x) \rightarrow P(x))$ and $S(a)$, obtain $P(a)$.

1. Put $x = a$ in the universal rule: $S(a) \rightarrow P(a)$.
2. Use the given fact $S(a)$.
3. Apply modus ponens to conclude $P(a)$.

Practice. From $\exists x(P(x) \wedge Q(x))$, show that $\exists xP(x)$.

Solution. Let c name one object for which $P(c) \wedge Q(c)$ holds. Then $P(c)$ holds. Since there is a specific example with P , we can conclude $\exists xP(x)$.

Practice. Do $\exists xP(x)$ and $\exists xQ(x)$ guarantee that the same object has both properties?

Solution. No. One student might have passed mathematics and another might have passed computing. Separate existence statements can have different examples.

Common error. One chosen example cannot prove an every statement. A proof about every object must use an arbitrary object, not one selected because it makes the question easy.

30 Writing a short, clear exam proof

A proof explains why a statement holds for every case covered by the question. Keep the working focused on the statement to be proved. Start with the given information, use a definition or rule, and finish by naming the property you have established.

Definitions to know.

- An even integer has the form $2k$.
- An odd integer has the form $2k + 1$.
- $a \mid b$ means $b = ak$ for some integer k .

In each definition, k must be an integer. You are not assuming a particular value of k .

Worked example. Prove that the sum of two even integers is even.

Let the two integers be $2r$ and $2s$, where r, s are integers. Their sum is

$$2r + 2s = 2(r + s).$$

Since $r + s$ is an integer, this is twice an integer. Therefore the sum is even. These few lines form a complete proof.

Practice. Prove that an even integer plus an odd integer is odd.

Solution. Write the numbers as $2r$ and $2s + 1$. Their sum is $2(r + s) + 1$, which has the form of an odd integer. Therefore the sum is odd.

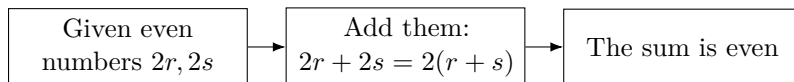
Practice. Why is checking $2 + 4 = 6$ not a complete proof about all even integers?

Solution. It checks only one pair. Writing $2r$ and $2s$ covers any pair of even integers.

Common error. Use different letters for independently chosen numbers. Writing both numbers as $2k$ would unintentionally make them equal. End by connecting your final expression to the required definition.

31 Direct proof from definitions

A direct proof of $P \rightarrow Q$ assumes P and derives Q . It is usually the first method to try when the hypothesis has a concrete algebraic or structural meaning.



Start from the definition and reach the required form.

Worked example. Prove that if $a \mid b$ and $b \mid c$ for integers a, b, c , then $a \mid c$. By the definition of divisibility, there are integers r, s such that $b = ar$ and $c = bs$. Substituting the first equality into the second gives

$$c = (ar)s = a(rs).$$

Since rs is an integer, this equality is exactly the definition of $a \mid c$. The proof does not need division by a , so it avoids unnecessary difficulties if $a = 0$.

This illustrates a reliable pattern: translate the hypotheses into usable equations, combine them, and identify the defining form of the conclusion. State why newly formed quantities still belong to the required domain.

Practice. Prove that the sum of an even integer and an odd integer is odd.

Solution. Let $a = 2r$ and $b = 2s + 1$, where $r, s \in \mathbb{Z}$. Then

$$a + b = 2r + 2s + 1 = 2(r + s) + 1.$$

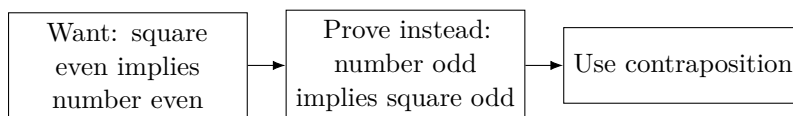
Because $r + s \in \mathbb{Z}$, the sum has the defining form of an odd integer. The arbitrary choices of a and b establish the universal claim, rather than only a particular numerical example.

Common error. Use separate names for independent witnesses. Writing two arbitrary even integers as $2k$ and $2k$ would accidentally make them equal. Write $2r$ and $2s$ unless equality is part of the assumptions.

The concluding sentence of a proof should connect the last expression to the exact claim. A correct calculation can still leave the reader uncertain if the property it establishes is not identified.

32 Proof by contraposition

Because $P \rightarrow Q$ is equivalent to $\neg Q \rightarrow \neg P$, an implication may be proved through its contrapositive. This is useful when the negation of the conclusion gives a simpler description than the original hypothesis.



The contrapositive is often easier to calculate.

Worked example. Prove: if n^2 is even, then n is even, for $n \in \mathbb{Z}$. The direct hypothesis $n^2 = 2k$ does not immediately describe n . Instead prove: if n is odd, then n^2 is odd. Write $n = 2r + 1$. Then

$$n^2 = 4r^2 + 4r + 1 = 2(2r^2 + 2r) + 1.$$

The bracketed quantity is an integer, so n^2 is odd. This establishes the contrapositive and therefore the original implication.

We used the fact that every integer is either even or odd. Over another domain, the phrase “not even” may not automatically mean “odd,” so domain information remains part of the reasoning.

Practice. Prove that if $a + b$ is odd, then at least one of the integers a, b is odd.

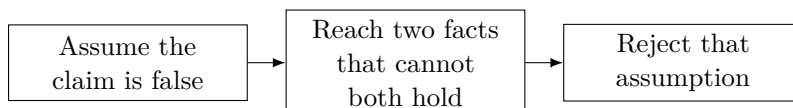
Solution. Negate the conclusion carefully. The negation of “at least one is odd” is “both are even.” Assume $a = 2r$ and $b = 2s$. Then $a + b = 2(r + s)$ is even, hence not odd. This proves the contrapositive. It does not claim that every pair with an odd member has odd sum; two odd members have an even sum.

Common error. Contraposition changes both the order and the signs. The converse $Q \rightarrow P$ is not an acceptable replacement for $P \rightarrow Q$. Write the intended contrapositive explicitly before beginning the algebra.

33 Proof by contradiction

To prove a statement S by contradiction, assume $\neg S$ and derive an impossibility using valid reasoning. The contradiction might be an equation such as $0 = 1$, a statement together with its negation, or a violation of a known theorem.

For a conditional theorem $P \rightarrow Q$, its negation is $P \wedge \neg Q$. Thus a contradiction proof assumes the hypothesis and denies the conclusion. The original assumptions remain active throughout.



A contradiction must be an actual impossibility.

Worked example. Prove that $\sqrt{2}$ is irrational. Suppose instead that $\sqrt{2} = a/b$ for integers a, b in lowest terms with $b \neq 0$. Squaring gives $a^2 = 2b^2$. Thus a^2 is even, so a is even by the preceding section. Write $a = 2k$. Substitution gives $4k^2 = 2b^2$, hence $b^2 = 2k^2$, so b is also even. Both numerator and denominator are divisible by 2, contradicting the choice of lowest terms. Therefore no such rational representation exists.

Practice. Prove that there is no largest integer.

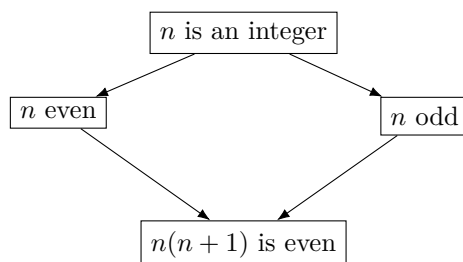
Solution. Suppose a largest integer M exists. The integer $M + 1$ is greater than M , contradicting its supposed maximality. Therefore the assumption of a largest integer is false. The proof uses closure of integers under adding 1 and the order property $M + 1 > M$.

Common error. A surprising consequence is not necessarily a contradiction. State the incompatible facts explicitly. Also distinguish contradiction from contraposition: contraposition proves another implication directly, while contradiction assumes the negation of the whole desired claim and rules it out.

Use contradiction when denial creates a useful object, such as a smallest counterexample, a rational representation, or a supposed extremal element. If a direct proof is shorter and clearer, there is no need to introduce an indirect argument.

34 Proof by cases

A proof by cases divides the possibilities into a collection that covers the whole domain, proves the conclusion in each case, and then combines the results. The cases need not always be disjoint, but they must be exhaustive.



Cover every possible case.

Worked example. Prove that $n(n+1)$ is even for every integer n . If n is even, the product has an even factor. If n is odd, $n+1$ is even, so the product again has an even factor. Every integer is either even or odd, so these two cases complete the proof.

A more algebraic presentation writes $n = 2k$ in the first case and $n = 2k + 1$ in the second. Then the products become $2k(2k+1)$ and $(2k+1)2(k+1)$, both visibly divisible by 2.

Practice. Prove that n^2 leaves remainder 0 or 1 on division by 3.

Solution. Every integer has one of the forms $3k$, $3k+1$, or $3k+2$. Squaring gives

$$\begin{aligned}(3k)^2 &= 3(3k^2), \\ (3k+1)^2 &= 3(3k^2+2k)+1, \\ (3k+2)^2 &= 3(3k^2+4k+1)+1.\end{aligned}$$

Thus the remainder is 0 in the first case and 1 in the other two. The division algorithm guarantees that no integer lies outside these cases.

Common error. When dividing into positive and negative cases, remember zero. When dividing by a modulus m , include all remainders $0, 1, \dots, m-1$. An omitted boundary case can invalidate an otherwise correct proof.

Good case divisions arise from the structure of the problem. Parity suggests two cases; division by 3 suggests three; an absolute value suggests cases according to sign.

35 Biconditional, existence, and uniqueness proofs

A biconditional $P \leftrightarrow Q$ requires two implications. An existence-and-uniqueness theorem requires two different tasks: construct or establish a witness, then show that any other witness must equal it. Keeping the obligations separate makes omissions easy to detect.

Worked example. Prove that n is even if and only if n^2 is even. In the forward direction, write $n = 2k$, giving $n^2 = 2(2k^2)$. In the reverse direction, use the contrapositive proof that an odd integer has an odd square. The two directions use different techniques, which is entirely acceptable.

For uniqueness, the standard method is to suppose u and v both satisfy the defining conditions and derive $u = v$. This does not assume that two distinct solutions exist; it shows that two candidate solutions cannot be distinct.

Practice. Prove that for each real a , there is exactly one real x satisfying $2x + 3 = a$.

Solution. For existence, choose $x = (a-3)/2$. Substituting gives $2((a-3)/2) + 3 = a$, so the proposed real number works. For uniqueness, suppose u and v both satisfy the equation. Then $2u + 3 = a = 2v + 3$. Subtracting 3 and dividing by the nonzero number 2 gives $u = v$. Thus the solution exists and is unique.

Common error. Finding a solution proves existence, not uniqueness. Conversely, showing that any two solutions must agree does not prove that a solution exists. An equation with no solutions also satisfies the condition “any two solutions are equal” vacuously.

When a theorem includes “exactly,” “only,” or “if and only if,” underline those words during planning. They usually signal an additional proof obligation beyond a single construction or implication.

36 Counterexamples and the limits of examples

A universal statement is refuted by one counterexample. An existential statement is established by one witness. These are different logical uses of a single example. Neither task requires an extensive list once a correct example is found.

For a conditional universal claim, a counterexample must satisfy the hypothesis and fail the conclusion. A number outside the hypothesis says nothing about whether the implication is universally true.

Worked example. The statement “For every integer n , if n is odd then n is prime” is refuted by $n = 9$. It is odd and not prime. The value 8 is not a counterexample, because it does not satisfy the hypothesis.

To refute a claimed equivalence, it is enough to find an interpretation where one side is true and the other false. To refute an argument, satisfy every premise while falsifying the conclusion. These are three versions of the same underlying discipline: target the exact logical structure.

Practice. A conjecture says that $n^2 + n + 41$ is prime for every nonnegative integer n . Find a counterexample without trying many unrelated values.

Solution. Choose $n = 41$. Then

$$41^2 + 41 + 41 = 41(41 + 2) = 41 \cdot 43,$$

which is composite. The constant term suggests choosing a multiple of 41 so that all terms share the factor. This does not explain every value of the polynomial, but one permitted failure is enough to disprove the universal conjecture.

Common error. A conjecture surviving hundreds of checks has evidence in its favour, not a proof over an infinite domain. On a finite domain, exhaustive checking can be a proof, but the exhaustiveness itself must be established.

Counterexamples are also diagnostic tools. After finding one, ask which assumption could be added or which conclusion weakened to obtain a true and useful revised statement.

Practice bank: Inference and short proofs

Attempt these questions before reading the answers on the following page. For a calculation or proof, write the method and the essential working as well as the final result.

1. From $p \vee q$, $\neg p$, and $q \rightarrow r$, derive r .
2. Prove that twice an odd integer is even.
3. Prove that two odd integers have an even sum.
4. Prove that if $3 \mid n$, then $3 \mid n^2$.
5. Disprove: every odd positive integer is prime.
6. State the contrapositive of: if n^2 is even, then n is even.
7. Prove that for every real a , the equation $x + 5 = a$ has one solution.
8. From $\forall x(P(x) \rightarrow Q(x))$ and $P(b)$, derive $Q(b)$.

Answers: Inference and short proofs

Compare the reasoning as well as the final answer. If your answer differs, return to the first step where the methods separate.

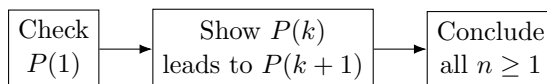
1. The first two statements give q by disjunctive syllogism. Then q and $q \rightarrow r$ give r by modus ponens.
2. Write the odd integer as $2k + 1$. Twice it is $4k + 2 = 2(2k + 1)$, which is twice an integer and therefore even.
3. Write them as $2r + 1$ and $2s + 1$. Their sum is $2(r + s + 1)$, which is even.
4. Write $n = 3k$. Then $n^2 = 9k^2 = 3(3k^2)$, so it is divisible by three.
5. Use 9. It is odd and positive but equals $3 \cdot 3$, so it is not prime.
6. If n is odd, then n^2 is odd, for integer n . Negate both parts and reverse their order.
7. Choose $x = a - 5$ to show existence. If $u + 5 = a$ and $v + 5 = a$, subtract five to obtain $u = v$, showing uniqueness.
8. Replace x by b in the universal rule, then use modus ponens with the fact $P(b)$.

37 Induction: the answer format to use

Mathematical induction is a method for proving a statement for $n = 1, 2, 3, \dots$, or for all integers beginning at another stated starting value. It has three main parts: check the first case, assume the case $n = k$, and prove the case $n = k + 1$.

Write your proof in this order.

1. State $P(n)$, the formula or claim you want to prove.
2. **Base case:** put in the first allowed value of n and check both sides.
3. **Induction hypothesis:** assume the formula holds at $n = k$.
4. **Induction step:** use that assumption to establish the formula at $n = k + 1$.
5. Conclude that the result holds for every allowed n .



Worked example. For $1 + 2 + \dots + n = n(n + 1)/2$, the induction hypothesis is $1 + \dots + k = k(k + 1)/2$. To reach the next sum, add the next term, $k + 1$. The next page shows the full calculation.

Practice. If the claim is stated for $n \geq 4$, which base case should you check?

Solution. Check $n = 4$. Then the step must work for each $k \geq 4$.

Practice. Is checking the first three values enough to complete induction?

Solution. No. It shows only three cases. You still need the step from an arbitrary k to $k + 1$.

Common error. Do not use the unproved formula for $k + 1$ as an assumption. Start from its left side and use the formula for k , which is the permitted hypothesis.

38 Induction for a sum formula

Consider the identity

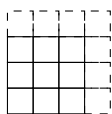
$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \quad (n \geq 1).$$

The notation on the left adds every integer from 1 through n . It can also be written $\sum_{i=1}^n i$, where i is a temporary summation index.

Worked example. For $n = 1$, both sides equal 1. Assume for an arbitrary $k \geq 1$ that $1 + \cdots + k = k(k+1)/2$. The next sum contains all the old terms plus one new term:

$$\begin{aligned} 1 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

The first equality is where the induction hypothesis is used. The final expression is exactly the claimed formula with n replaced by $k+1$.



A 3×3 square
plus seven squares
makes a 4×4 square.

The next odd number increases k^2 to $(k+1)^2$.

Practice. Prove $1 + 3 + 5 + \cdots + (2n-1) = n^2$ for $n \geq 1$.

Solution. The base case is $1 = 1^2$. Assume the sum of the first k odd positive integers is k^2 . The next odd number is $2(k+1) - 1 = 2k+1$, so the next sum is

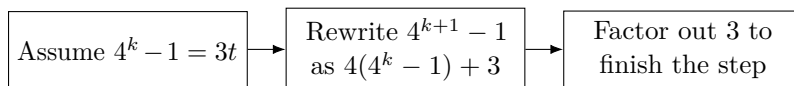
$$k^2 + (2k+1) = (k+1)^2.$$

This proves the inductive step and hence the identity for all positive n . The calculation is short because adding the next odd number supplies exactly the amount needed to grow a square of side k to one of side $k+1$.

Common error. Keep the “next term” separate from the old sum before substituting the hypothesis. Replacing the entire $(k+1)$ -term sum by the unproved formula would assume the desired result.

39 Induction for divisibility

Divisibility problems often become manageable when the next expression is rewritten as a multiple of the previous expression plus a known divisible quantity. This creates a place to use the induction hypothesis.



Expose the expression already covered by the hypothesis.

Worked example. Prove $3 \mid (4^n - 1)$ for every integer $n \geq 0$. At $n = 0$, the expression is 0, which is divisible by 3. Assume $4^k - 1 = 3m$ for some integer m . Then

$$\begin{aligned} 4^{k+1} - 1 &= 4(4^k - 1) + 3 \\ &= 4(3m) + 3 = 3(4m + 1). \end{aligned}$$

Since $4m + 1$ is an integer, the next expression is divisible by 3. This proves the step.

The useful decomposition was not $4 \cdot 4^k - 1$ alone. Adding and subtracting the appropriate constant exposed the expression $4^k - 1$ that the hypothesis already controls.

Practice. Prove $6 \mid (7^n - 1)$ for every integer $n \geq 0$.

Solution. At $n = 0$, $7^0 - 1 = 0$. Assume $7^k - 1 = 6m$. Then

$$\begin{aligned} 7^{k+1} - 1 &= 7(7^k - 1) + 6 \\ &= 7 \cdot 6m + 6 \\ &= 6(7m + 1). \end{aligned}$$

Thus the next value is also divisible by 6. The argument establishes the result for every nonnegative integer by induction.

Common error. Saying “the expression is clearly divisible” hides the key step. Identify an integer multiplier or cite a previously justified divisibility rule. If using congruence notation, define $a \equiv b \pmod{m}$ to mean that m divides $a - b$; it is not the same use of $\equiv$ as logical equivalence, though the context distinguishes them.

After learning the inductive proof, notice the shorter modular explanation $7 \equiv 1 \pmod{6}$. Different methods can establish the same fact; the inductive version specifically practises how a property is preserved from one index to the next.

40 Induction for inequalities

An inductive inequality proof must preserve the direction of every comparison. Multiplication by a positive quantity preserves the direction; multiplication by a negative quantity reverses it. The starting index is often determined by where an auxiliary inequality becomes valid.

Worked example. Prove $2^n \geq n + 1$ for $n \geq 0$. At $n = 0$, both sides equal 1. Assume $2^k \geq k + 1$. Multiplying by 2 gives

$$2^{k+1} \geq 2(k + 1) \geq k + 2,$$

where the second inequality holds because $k \geq 0$. The right side is $(k + 1) + 1$, so the desired next case follows.

Practice. Prove $2^n \geq n^2$ for all integers $n \geq 4$.

Solution. At $n = 4$, both sides equal 16. Assume $2^k \geq k^2$ for some $k \geq 4$. Then $2^{k+1} \geq 2k^2$. We need $2k^2 \geq (k + 1)^2$, which is equivalent to

$$k^2 - 2k - 1 \geq 0.$$

For $k \geq 4$, $k^2 - 2k - 1 = (k - 1)^2 - 2 \geq 9 - 2 > 0$. Therefore $2^{k+1} \geq (k + 1)^2$, completing the step.

Common error. The inequality is false at $n = 3$, since $8 < 9$. Starting at $n = 0$ and checking only that the initial equality holds would not solve the problem: the intended successor step must also work at every intermediate index. Induction cannot jump over a failed case without an explicit change in its structure.

When planning, write the target expression for $k + 1$ before manipulating the hypothesis. This makes clear which numerical comparison must bridge the old bound and the new one.

41 Strong induction in simple terms

Ordinary induction uses the previous case. Strong induction allows you to use any earlier cases that have already been covered. Use it when the next answer depends on more than one earlier answer, or when a number is broken into smaller factors.

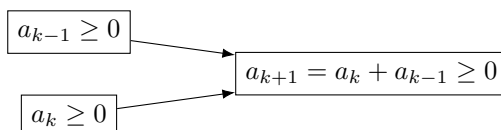
Answer pattern. Check the starting cases. Assume the claim holds for all required values up to k . Prove it for $k + 1$ using the earlier cases you need. Then state the conclusion.

Worked example. Let $a_0 = 0$, $a_1 = 1$, and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$. Prove that all the terms are nonnegative.

The starting values 0 and 1 are nonnegative. Assume the earlier terms through a_k are nonnegative, with $k \geq 1$. Then

$$a_{k+1} = a_k + a_{k-1}$$

is the sum of two nonnegative numbers, so it is nonnegative. This completes the induction.



Practice. Why were two starting values checked?

Solution. The recurrence uses two earlier terms. Both must be available before the first recurrence step at $n = 2$.

Practice. In a factorisation proof for $k + 1 = ab$, why might strong induction help?

Solution. The factors a, b may be much smaller than k , rather than equal to k . Assuming the result for all earlier allowed integers lets you use it for both factors.

Common error. Strong induction still needs a base case and a complete step. The stronger assumption makes some proofs easier; it does not remove either requirement.

42 A second full induction example

For an exam, practise at least one complete induction proof without looking at a template. The following example uses a sum of powers and only one algebraic step in the induction stage.

Worked example. Prove

$$1 + 2 + 4 + \cdots + 2^n = 2^{n+1} - 1 \quad (n \geq 0).$$

Base case. At $n = 0$, the left side contains just 1. The right side is $2^1 - 1 = 1$. So the first case holds.

Hypothesis. Assume

$$1 + 2 + \cdots + 2^k = 2^{k+1} - 1.$$

Next case. Add the next term, 2^{k+1} :

$$\begin{aligned} 1 + 2 + \cdots + 2^k + 2^{k+1} &= (2^{k+1} - 1) + 2^{k+1} \\ &= 2 \cdot 2^{k+1} - 1 \\ &= 2^{k+2} - 1. \end{aligned}$$

This is the required formula with $n = k + 1$. Therefore the result holds for all $n \geq 0$.

n	Sum	$2^{n+1} - 1$
0	1	1
1	$1 + 2 = 3$	3
2	$1 + 2 + 4 = 7$	7

Practice. What is the new term when passing from index k to $k + 1$ in this example?

Solution. It is 2^{k+1} . The old sum already ends at 2^k .

Practice. Use the formula to find $1 + 2 + 4 + \cdots + 32$.

Solution. Since $32 = 2^5$, use $n = 5$. The sum is $2^6 - 1 = 63$.

Common error. This sum starts at 2^0 , so its starting index is zero. Check the first and last terms before substituting into the formula.

43 Finding the mistake in an induction proof

An exam may give an attempted proof and ask you to identify its error. Check the base case, the size of the step, the use of the hypothesis, and the algebra. Point to the exact line that fails.

Worked example. A student claims that every positive integer is odd. They check that 1 is odd, then argue that adding 2 to an odd integer gives another odd integer. The calculation is correct, but it proves only the cases $1, 3, 5, \dots$. It never reaches $2, 4, 6, \dots$

The missing link is the step from k to $k + 1$. A step from k to $k + 2$ advances along only one of the two index sequences unless both starting cases are checked.

A practical checking order.

1. Test the first allowed value and the next value.
2. Check that the claimed step reaches every required index.
3. Locate the line that uses the induction hypothesis.
4. Expand the final expression yourself to check the algebra.

Practice. A student claims $1 + \dots + n = n^2$. Their next-case calculation is $k^2 + k + 1 = (k + 1)^2$. What is wrong?

Solution. The equality is false. Expanding the right side gives $k^2 + 2k + 1$, which contains an extra k . Also, the claimed formula fails at $n = 2$: the sum is 3, not 4. A correct base case at $n = 1$ cannot rescue a false induction step.

Practice. A proof checks $P(3)$ and establishes $P(k) \rightarrow P(k + 1)$ only for $k \geq 5$. Does it prove all cases from 3 onward?

Solution. No. The transitions from 3 to 4 and from 4 to 5 have not been covered. Establish those cases separately, or show that the step also works at $k = 3, 4$.

Common error. Writing the labels base case and induction step does not make the reasoning valid. Each labelled part must perform its stated job.

44 Choosing a proof technique

The logical shape of a claim suggests a starting method. For a universal conditional, assume an arbitrary object satisfies the hypothesis. For existence, try to construct a witness. For uniqueness, compare two possible witnesses. For a sequence of assertions indexed by integers, look for an inductive relation.

Use contraposition when a negated conclusion is easy to describe. Use contradiction when a denied claim creates a useful object, such as a smallest counterexample. Use cases when the domain naturally separates by parity, sign, or remainder. These are choices guided by structure, not rigid rules.

Worked example. Claim: if the product of two integers is odd, both integers are odd. The contrapositive is especially natural: if at least one factor is even, the product is even. If $a = 2k$, then $ab = 2(kb)$; the same reasoning applies if b is even. This proves the contrapositive and hence the claim.

Practice. Select and carry out a short method for: (a) there exists an even prime; (b) no integer is both even and odd; (c) if $a \mid b$, then $a \mid bc$ for integer c .

Solution. (a) A witness is 2: it is even and has exactly two positive divisors, 1 and 2. (b) Suppose $n = 2r = 2s + 1$. Then $2(r - s) = 1$, impossible for integers because an even integer cannot equal 1; equivalently, $r - s = 1/2$ contradicts integrality. (c) Directly, write $b = ak$. Then $bc = a(kc)$, and kc is an integer.

Common error. Do not choose induction merely because a statement mentions integers. If its proof does not depend on moving from smaller values to larger ones, a direct argument may be simpler and more informative.

Practice bank: Mathematical induction

Attempt these questions before reading the answers on the following page. For a calculation or proof, write the method and the essential working as well as the final result.

1. For $1 + \cdots + n = n(n+1)/2$, write the base-case check.
2. For the same formula, write the induction hypothesis.
3. Complete the step for that sum.
4. Prove $2 + 4 + \cdots + 2n = n(n+1)$ by induction.
5. What term is added when extending $1 + 3 + \cdots + (2k-1)$ to the next case?
6. Prove $5 \mid (6^n - 1)$ for $n \geq 0$ by induction.
7. A proof checks $P(1)$ and shows $P(k) \rightarrow P(k+2)$. What may be missing?
8. Complete an induction step for $2^n \geq n+1$.

Answers: Mathematical induction

Compare the reasoning as well as the final answer. If your answer differs, return to the first step where the methods separate.

1. At $n = 1$, the left side is 1 and the right side is $1(2)/2 = 1$.
2. Assume $1 + \cdots + k = k(k+1)/2$ for an arbitrary allowed integer $k \geq 1$.
3. Add $k+1$ to the old sum: $k(k+1)/2 + (k+1) = (k+1)(k+2)/2$. This is the formula at $k+1$.
4. The base case is $2 = 1(2)$. Assume the result at k . Add $2(k+1)$ to get $k(k+1) + 2(k+1) = (k+1)(k+2)$, the next case.
5. The next odd term is $2(k+1) - 1 = 2k+1$. Thus the new sum is the old sum plus $2k+1$.
6. At zero the expression is zero. Assume $6^k - 1 = 5t$. Then $6^{k+1} - 1 = 6(6^k - 1) + 5 = 5(6t + 1)$, so the next case holds.
7. The even-indexed cases. Add a base case $P(2)$, or prove a step of size one.
8. Assume $2^k \geq k+1$. Then $2^{k+1} \geq 2k+2 \geq k+2$ for $k \geq 0$. Also check the base case $n = 0$.

45 Resolution: cancel a matching opposite pair

Resolution works with OR clauses. If one clause contains p and another contains $\neg p$, cross out that pair and join the remaining parts with OR:

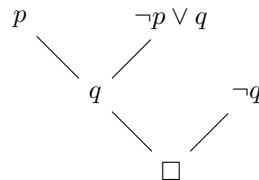
$$(p \vee q), (\neg p \vee r) \implies q \vee r.$$

This is an inference rule, not ordinary cancellation within a single expression.

To prove a conclusion by resolution:

1. Convert the premises into clauses.
2. Add the negation of the conclusion.
3. Resolve clauses containing matching opposite literals.
4. Continue until you obtain the empty clause $\square$, meaning a contradiction.

Worked example. Prove q from p and $p \rightarrow q$. The implication becomes $\neg p \vee q$. Add $\neg q$, the negation of the conclusion. Resolve p with $\neg p \vee q$ to get q . Resolve q with $\neg q$ to get $\square$. Therefore the conclusion follows.



Practice. Resolve $p \vee q$ and $\neg p \vee s$.

Solution. The result is $q \vee s$. The remaining alternatives are joined by OR, not AND.

Practice. Resolve r and $\neg r$.

Solution. No literals remain, so the result is $\square$.

Common error. Do not delete both p and $\neg p$ from the single clause $p \vee \neg p$. That clause is always true. The resolution rule uses two premise clauses.

46 Matching variables before using resolution

In predicate questions, a variable may need to be replaced by a name before two literals match. This matching step is called unification. The replacement is called a substitution.

Worked example. A rule says that every registered person is eligible:

$$\neg R(x) \vee E(x).$$

You also know $R(\text{Mira})$. To match this fact with $R(x)$, replace x by Mira everywhere in the rule:

$$\neg R(\text{Mira}) \vee E(\text{Mira}).$$

Resolve with the fact $R(\text{Mira})$. The remaining statement is $E(\text{Mira})$.

Working format. Write the replacement first, such as $x \mapsto \text{Mira}$, then write the resulting clause. The arrow $\mapsto$ means replace the variable by the expression on its right.

Practice. Match $P(x, 5)$ with $P(2, y)$.

Solution. Use $x \mapsto 2$ and $y \mapsto 5$. Both expressions then become $P(2, 5)$.

Practice. Can $P(x, x)$ be matched with $P(2, 3)$?

Solution. No. The first position would require $x = 2$, while the second would require $x = 3$. A repeated variable must receive the same replacement everywhere.

Practice. From $\neg S(x) \vee P(x)$ and $S(a)$, what follows?

Solution. Replace x by a in the whole clause. Then resolve $\neg S(a) \vee P(a)$ with $S(a)$ to get $P(a)$.

Common error. Do not change only the literal you are crossing out. Apply the substitution to the remaining literals too. This is how the correct answer name is carried into the result.

47 Preparing a statement for predicate resolution

Predicate resolution needs clauses: statements written as ORs of literals. For the basic rule questions, the main task is removing implication arrows.

Worked example. Convert every student is eligible:

$$\forall x(S(x) \rightarrow E(x)).$$

Replace the implication with OR and NOT:

$$\neg S(x) \vee E(x).$$

The variable is understood to be universally quantified in this clause form.

Worked example. Convert $P(x) \wedge Q(x) \rightarrow R(x)$. Negate the whole left side when removing the arrow:

$$\neg(P(x) \wedge Q(x)) \vee R(x) \equiv \neg P(x) \vee \neg Q(x) \vee R(x).$$

If an existential statement appears. An assertion such as $\exists yP(y)$ says that some suitable object exists. Give it a fresh name c , producing $P(c)$. Do not reuse a name already given to another object.

For $\forall x\exists yR(x, y)$, the choice of y may depend on x , so use $R(x, f(x))$. The notation $f(x)$ records a suitable choice for that particular x . This step is called Skolemisation; its purpose here is preparing clauses while preserving whether the statements can all be satisfied.

Practice. Convert $A(x) \rightarrow (B(x) \wedge C(x))$ into clauses.

Solution. Remove the arrow and distribute OR over AND. The two clauses are $\neg A(x) \vee B(x)$ and $\neg A(x) \vee C(x)$.

Practice. In $\exists y\forall xR(x, y)$, should the replacement be one name or a choice depending on x ?

Solution. Use one fresh name c . The statement asks for the same y to work for every x .

Common error. Keep the quantifier order when choosing a replacement. It tells you whether one fixed name is enough.

48 Answer extraction: find the value, not just yes

An ordinary proof may show that someone is eligible. Answer extraction also reports who. In basic resolution questions, keep track of the variable replacements used in the successful proof.

Worked example. The facts and rule are

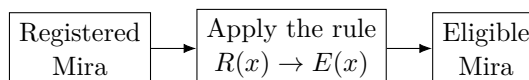
$$R(\text{Mira}), \quad R(x) \rightarrow E(x).$$

The question asks for a value of x for which $E(x)$ holds.

Simple working.

1. The rule says registered implies eligible.
2. The fact says Mira is registered.
3. Replace x by Mira and conclude $E(\text{Mira})$.
4. Report the answer: $x = \text{Mira}$.

If the question asks for an answer literal, use $\text{Answer}(x)$ as a label that carries the answer. Start with $\neg E(x) \vee \text{Answer}(x)$ and the rule clause $\neg R(y) \vee E(y)$. Resolve to obtain $\neg R(y) \vee \text{Answer}(y)$. The fact $R(\text{Mira})$ then gives $\text{Answer}(\text{Mira})$.



Practice. If $R(\text{Arun})$ is also given, what answers follow?

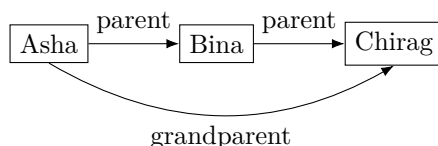
Solution. Both Mira and Arun. Apply the same rule separately to each fact.

Common error. If the question asks for all answers, do not stop after finding the first one. Report the replacements supported by the facts and rules; do not invent additional people.

49 Answer extraction through two linked facts

Some queries need a chain of facts. Follow the chain while keeping the shared name the same at each step.

Worked example. Suppose Asha is a parent of Bina, and Bina is a parent of Chirag. Who is a grandparent of Chirag?



Let $P(x, y)$ mean that x is a parent of y . The rule is

$$P(x, y) \wedge P(y, z) \rightarrow G(x, z).$$

The repeated y links the two facts. For the query $G(x, \text{Chirag})$, use $z = \text{Chirag}$. The second parent fact gives $y = \text{Bina}$. The first then gives $x = \text{Asha}$. Thus the answer is Asha.

Resolution working, if required. Begin with the answer clause $\neg G(w, \text{Chirag}) \vee \text{Answer}(w)$. Resolving with the rule gives

$$\neg P(x, y) \vee \neg P(y, \text{Chirag}) \vee \text{Answer}(x).$$

Use the two parent facts in turn. The final result is $\text{Answer}(\text{Asha})$.

Practice. If Dev is also a parent of Bina, who else can be reported?

Solution. Dev, because the chain Dev–Bina–Chirag also fits the rule.

Practice. Can a parent fact about an unrelated person replace the link through Bina?

Solution. No. The same middle person must occur in both parent relationships.

Common error. Missing facts do not automatically mean a relationship is false. For these questions, report the answers that can be obtained from the supplied information.

50 Unit 2 exam method sheet

Use the wording of the question to choose your first step. Keep the working short enough to follow, but show the part that justifies your answer.

Question asks for	Start this way
Truth table	List every true/false combination; add intermediate columns.
Tautology or contradiction	Read the final column: all true or all false.
Equivalence	Compare final columns or apply named logical laws.
Validity	Look for true premises with a false conclusion.
CNF or DNF	Remove arrows, push NOT inward, expand into the required shape.
Quantifier translation	Identify the domain and the order of every/some.
Negation	Switch quantifiers and negate the inner statement.
Induction	Base case, hypothesis at k , step to $k + 1$, conclusion.
Answer extraction	Follow the rules and record the variable replacements.

Practice. How do you disprove a universal statement?

Solution. Give one allowed example for which it fails. If it is an implication, the example must satisfy the hypothesis and fail the conclusion.

Practice. How do you prove an existential statement?

Solution. Give a suitable example and check that it satisfies the required property.

Practice. What does a biconditional proof need?

Solution. Both directions: prove $P \rightarrow Q$ and $Q \rightarrow P$.

Final answer check. Have you answered the exact question? A satisfiable formula need not be a tautology. A valid argument need not have true premises. An example does not prove an every statement. An answer to a some statement need not be unique.

The next practice set mixes the topics. Try identifying the method before carrying out the calculation; that choice is often the most important step.

Practice bank: Resolution and mixed revision

Attempt these questions before reading the answers on the following page. For a calculation or proof, write the method and the essential working as well as the final result.

1. Resolve $a \vee b$ with $\neg a$.
2. Resolve $\neg p \vee q$ with $p \vee r$.
3. What does the empty clause mean?
4. Match $R(x, 2)$ with $R(5, y)$.
5. From $S(a)$ and $S(x) \rightarrow E(x)$, extract an answer to $E(z)$.
6. If $S(b)$ is added to the previous facts, what answers can you give?
7. Write the clause for $P(x) \wedge Q(x) \rightarrow R(x)$.
8. Prove q by refutation from p and $\neg p \vee q$.

Answers: Resolution and mixed revision

Compare the reasoning as well as the final answer. If your answer differs, return to the first step where the methods separate.

1. The matching opposite pair is $a, \neg a$. The remaining clause is b .
2. The result is $q \vee r$. Keep the remaining alternatives joined by OR.
3. It means falsehood or a contradiction. In a refutation proof, deriving it shows that the premises and the negated conclusion cannot all hold.
4. Use $x \mapsto 5$ and $y \mapsto 2$. Both expressions become $R(5, 2)$.
5. Apply the rule with $x = a$ to get $E(a)$. Report the query answer $z = a$.
6. Both $z = a$ and $z = b$, because the rule applies to each fact.
7. The clause is $\neg P(x) \vee \neg Q(x) \vee R(x)$. Negating the AND on the left produces two negated alternatives.
8. Add $\neg q$. Resolve the two original premises to get q , then resolve q with $\neg q$ to get the empty clause.

51 Cumulative practice questions

Attempt these questions without the solutions in the next section. For every symbolic answer, include a sentence explaining what it means. For a false universal claim or invalid argument, provide a specific counterexample or countermodel.

1. Simplify $\neg(p \rightarrow q) \vee (p \wedge q)$.
2. Find a CNF equivalent to $(p \wedge q) \rightarrow r$. Is it a tautology?
3. Decide whether $p \rightarrow q$, $q \rightarrow r$, and $\neg r$ entail $\neg p$. Give a short derivation.
4. Over integers, compare the truth of $\forall x \exists y (x + y = 0)$ and $\exists y \forall x (x + y = 0)$.
5. Negate “Every student has read some book” using separate domains of students and books. Explain the resulting sentence.
6. Translate “Exactly one student solved every problem.” Use $S(x)$ for student, $R(y)$ for problem, and $L(x, y)$ for solved, with a common domain of relevant objects.
7. Prove that the square of an odd integer leaves remainder 1 on division by 8.
8. Prove by induction that $1 + 2 + 4 + \cdots + 2^n = 2^{n+1} - 1$ for $n \geq 0$.
9. Use resolution to show that $p \vee q$, $\neg p \vee r$, and $\neg q \vee r$ entail r .
10. Given $A(a)$, $\forall x (A(x) \rightarrow B(x))$, and $\forall x (B(x) \rightarrow C(x))$, extract an answer to the query $C(z)$, showing the bindings.

Suggested working method. For Questions 1–3 and 9, identify the connectives before applying a law or rule. For Questions 4–6, state the permitted dependence of each witness. For Questions 7–8, write the domain, the definition or induction hypothesis, and the precise final form needed. For Question 10, carry the answer variable through the proof instead of simply stating that an answer exists.

Self-check. A complete solution should reveal why it works. The final formula alone is insufficient for a proof question, while several pages of calculation are unnecessary for a counterexample question once the relevant failure is explicitly demonstrated.

52 Cumulative solutions and further reading

Solutions 1–5

1. Since $\neg(p \rightarrow q) \equiv p \wedge \neg q$, the expression becomes $(p \wedge \neg q) \vee (p \wedge q) \equiv p \wedge (\neg q \vee q) \equiv p$.
2. Removing the implication and applying De Morgan's law gives $\neg p \vee \neg q \vee r$. This is a single CNF clause. It is not a tautology: $p = q = \text{T}, r = \text{F}$ makes it false. It is satisfiable, for example whenever $r = \text{T}$.
3. From $q \rightarrow r$ and $\neg r$, modus tollens gives $\neg q$. From $p \rightarrow q$ and $\neg q$, another use of modus tollens gives $\neg p$. Thus the argument is valid.
4. The first statement is true: for each integer x , choose the integer $y = -x$. The second is false: $x = 0$ would require $y = 0$, while $x = 1$ would require $y = -1$. One fixed y cannot meet both demands.
5. If $L(x, y)$ means that student x read book y , the original is $\forall x \exists y L(x, y)$. Its negation is $\exists x \forall y \neg L(x, y)$: some student has read no book in the stated domain.

Solutions 6–10

6. Define $H(x)$ as an abbreviation for $S(x) \wedge \forall y(R(y) \rightarrow L(x, y))$. Then write $\exists x(H(x) \wedge \forall z(H(z) \rightarrow z = x))$. This permits other students to solve some problems, but exactly one solves every problem. The abbreviation is fully specified and can be expanded if required.
7. Write $n = 2k + 1$. Then $n^2 = 4k(k + 1) + 1$. One of the consecutive integers $k, k + 1$ is even, so $k(k + 1) = 2m$ for some integer m . Hence $n^2 = 8m + 1$, proving the remainder statement.
8. At $n = 0$, the sum is $1 = 2^1 - 1$. Assume the sum through 2^k equals $2^{k+1} - 1$. Adding the next term gives $(2^{k+1} - 1) + 2^{k+1} = 2^{k+2} - 1$, the required case at $k + 1$. Therefore induction applies for all $n \geq 0$.
9. Add $\neg r$. Resolving it with $\neg p \vee r$ yields $\neg p$, and with $\neg q \vee r$ yields $\neg q$. Resolve $p \vee q$ with $\neg p$ to get q , then with $\neg q$ to get $\square$. Thus premises plus the negated conclusion are unsatisfiable.
10. Start with $\neg C(z) \vee \text{Answer}(z)$. Resolve against $\neg B(y) \vee C(y)$ to obtain $\neg B(y) \vee \text{Answer}(y)$. Resolve against $\neg A(x) \vee B(x)$ to obtain $\neg A(x) \vee \text{Answer}(x)$. The fact $A(a)$ binds the remaining variable to a , giving $\text{Answer}(a)$. The answer substitution is $z \mapsto a$.

Further reading and scope

The explanations and exercises in these notes are original instructional material. For additional treatments, consult Eric Lehman, F. Thomson Leighton, and Albert R. Meyer, *Mathematics for Computer Science*, available through MIT OpenCourseWare. Its logic and proof chapters provide a broader discrete-mathematics context.

For the resolution-based meaning of answer extraction, see the Stanford Introduction to Logic supplement on resolution and the University of Texas notes on answer extraction. Here answer extraction is developed with explicit substitutions and definite-rule examples; general automated theorem proving has additional search and completeness issues.